

Gravity development board

Designed to help reduce the cost of space-tolerant electronics, Blaze Sanders designed a board to withstand the harsh environments of space. Some claim this board to be

a good alternative for Arduino Uno, which has been the other preferred prototyping environment. The gravity development board (GDB) started life as the high-power driver for Lunar Wombat, which its creator was

working on for NASA. "We needed a small embedded system that could drive Firgelli L16 linear actuators and act as the control and communication system for the entire Lunar Wombat," he states in an article about the board.

Some of the challenges that face an electronic board designed for space include its ability to handle the vibrations of launch vehicles, heat distribution in a weightless atmosphere (convention does not work in space) and redundancies to prevent single-event upsets caused by radiation.

The GDB is designed for beginners and experts, and can be programmed in four languages, including 12-blocks GUI, SPIN, C and assembly.

The board comes in three forms:

1. E series for Earth based applications where size and ease of development matter,
2. M series for Mars or planetary based applications and
3. S series for free space based applications where rad-hard components are a must. This type also comes with Xilinx Virtex-5Q field programmable gate arrays (FPGAs), allowing reconfigurable hardware.

Would you like to be RocketMan

Ever seen the movie *RocketMan* from 1997 where an engineer gets to ride a rocket to planet Mars? Well, you could start working for your chance at it too, by building some of these electronics projects designed to help control your rocketship.

Altus Metrum

Although they have a collection of open hardware and software designs available at their website, TeleMetrum and TeleMega are the ones at the top of the line. Powered by ST Micro's STM32L151 ARM Cortex M3 based microcontroller (MCU), these both come with TI's CC1120 RF transceivers, u-blox's MAX-7Q GPS receivers and Freescale's MMA6555 inertial sensor, among other components. These boards were designed using the open source gEDA (an open source

How to select a developer board

If one has to implement simple logic based algorithms using basic control loop structures of programming, a MCU based developer kit would be apt. "Arduino and its variants would be easy to learn and allow quick prototyping. Most MCUs have an embedded C based programming environment, although in recent times we have seen Java/.net cross compilers for MCUs. However, C based IDEs are stable and remain the most widely used," explains Darshan Virupaksha, systems engineer, Altix Innovations and co-organiser of IoTBLR.

Computing and power. He adds that if one has more computational requirements like image processing and intends to have faster communications interfaces, Raspberry Pi, Beaglebone or UDOO boards are useful. High-level languages are very well supported on any of these Linux platforms. Since these platforms run some flavour of Linux, prototyping becomes simpler. Hence, for use cases involving all media processing, machine learning algorithms and faster connectivity, one could use the above-mentioned Linux platforms.

All MCU based boards can easily run on battery power. Although Raspberry Pi can be technically supported with just a battery, it is not a viable option. While a MCU based board would support low power requirements (0.3W), the latter would need around 10W of continuous power. Hence it makes more sense to have a MCU based board, like Arduino in these cases.

Cost. Boards like Raspberry Pi are cheaper since their production numbers are quite high. This allows the company to minimise manufacturing cost due to economies of scale, and pass on the savings to the end customer. "It is also much easier to integrate an open source board rather than developing your own PCB from scratch," adds Upton.

"I started my son off with Snap Circuits, and he has moved on to Lego Mindstorms in his local robotics club at school. I would like to see accessibility improve with lower prices, and many boards from major semiconductor companies are already subsidised. But for the best marriage of accessible (that is, low cost) hardware and information, I think Arduino has that nailed," explains Lynette Reese, technical team lead, Mouser Electronics.

Reduced time to market. "Open source boards have been a great tool for prototyping and R&D, because the boards were well documented and as a result offered much better support compared to proprietary boards," says Eben.

He adds that engineers would also always want to figure out a way to get things done as quickly as possible. Additionally, even with the help of a board, there is still plenty of hard work left in the form of integration.

"Using such platforms is advantageous for product developers as the complete hardware design, firmware, open source OS and reference projects are already available and give them a head start in their product development. The developer will have to only focus on developing the application and ID design (look and feel) to take the product to market," suggests Selvaraj Kaliyappan, vice president - PES Engineering, Mistral Solutions.

Community. "I am a big fan of Arduino and it is kind of the original developer board. It is also good to see that they are diversifying into new kinds of boards," adds Eben. It is also very powerful because it has a very big community, a good flow and also helps with figuring out how to get Atmel chips integrated into your product. Arduino can help you get past these kinds of problems.

The community following is not limited to the Arduino's and Raspberry Pi's of the world though. There are numerous maker groups, organisations and communities doing their own things too. You can meet up with most of them at EFY's Electronics Rocks branded conferences in Bengaluru. The next one is scheduled for January 11-13, 2016, to be held concurrently with EFY Expo 2016 in Bengaluru.

Ganesh Kumar, country sales manager, Atmel, says, "Due to the nature of our MCUs being deployed in a wide spectrum of applications and industries, we have accustomed to servicing not just a few large customers, but long-tail customers, among them many are makers or start-ups. We believe that these currently small players will eventually play a key role in shaping the up-and-coming Internet Of Things (IoT) revolution. According to Garner, 50 per cent of the IoT solutions by 2018 will be provided by start-ups that are less than three years old. We believe that many such key IoT players will emerge from the makers."